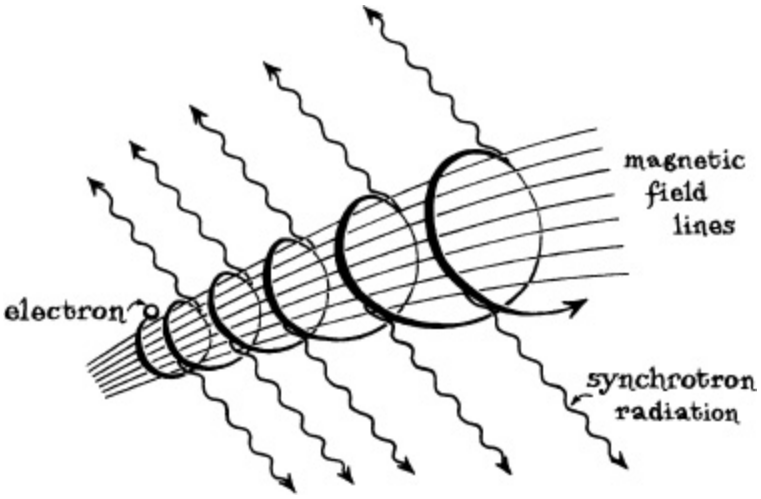
1940s, despite physicists' considerable interest in synchrotron radiation, astronomers paid no attention to it. The astronomers' mental block held sway.



9.4 Cosmic radio waves are produced by near-light-speed electrons that spiral around and around in magnetic fields. The magnetic field forces an electron to spiral instead of moving on a straight line, and the electron's spiraling motion produces the radio waves.

In 1950 Karl Otto Kiepenheuer in Chicago and Vitaly Lazarevich Ginzburg in Moscow (the same Ginzburg who had invented the LiD fuel for the Soviet hydrogen bomb, and who had discovered the first hint that black holes have no hair[2]) broke the mental block. Building on seminal ideas of Hans Alfven and Nicolai Herlofson, Kiepenheuer and Ginzburg proposed (correctly) that Jansky's radio waves from our own galaxy are synchrotron radiation produced by near-light-speed electrons spiraling around magnetic field lines that fill interstellar space (Figure 9.4).

A few years later, when the giant radio-emitting lobes of radio galaxies and then quasars were discovered, it was natural (and correct) to conclude that their radio waves were also produced by electrons spiraling around magnetic field lines. From the physical laws governing such spiraling and the properties of the observed radio waves, Geoffrey Burbidge at the University of California in San Diego computed how much energy the lobes' magnetic field and high-speed electrons must have. His startling answer: In the most extreme cases, the radio-emitting lobes must have about as much magnetic energy and high-speed *(kinetic)* energy as one would get by converting all the mass of 10 million (10^7) Suns into pure energy with 100 percent efficiency.